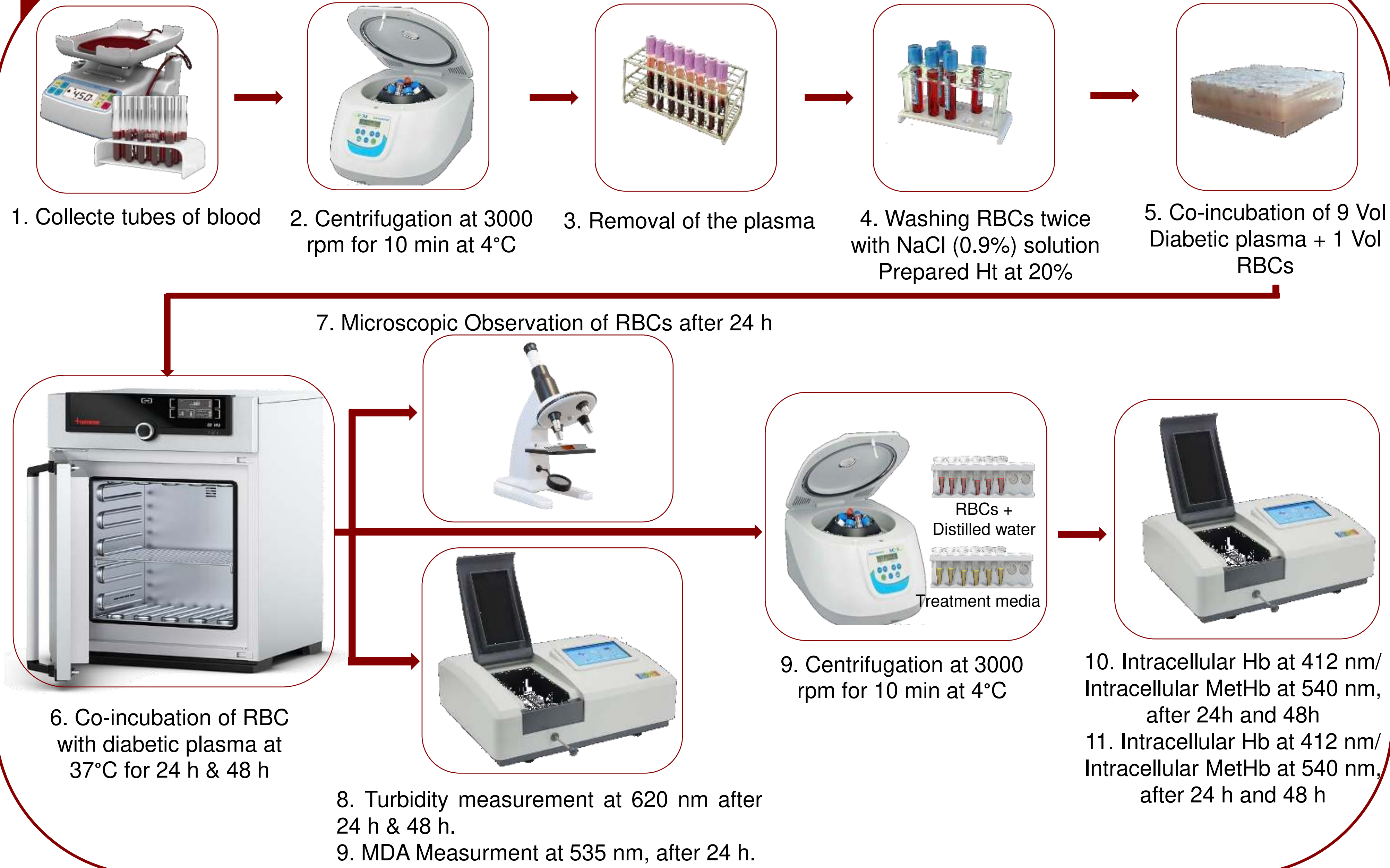


### Introduction

Hyperglycemia has traditionally been considered the primary trigger of diabetic complications. However, many patients live with prolonged diabetes without developing severe outcomes, suggesting the involvement of additional pathogenic mechanisms. Increasing evidence points to oxidative stress (OS) as a central contributor. Erythrocytes, which are continuously exposed to the diabetic plasma environment, are particularly vulnerable to oxidative damage. This vulnerability is associated with chronic hyperglycemia-induced non-enzymatic glycation of biomolecules, notably hemoglobin glycation, leading to alterations its structure and function. In this context, the present study was undertaken to explore the effect of diabetic plasma at different HbA1c levels on human RBCs in order to highlight the amplification mechanisms of OS in relation to metHb production, a potential bio-indicator that could be related to diabetes.

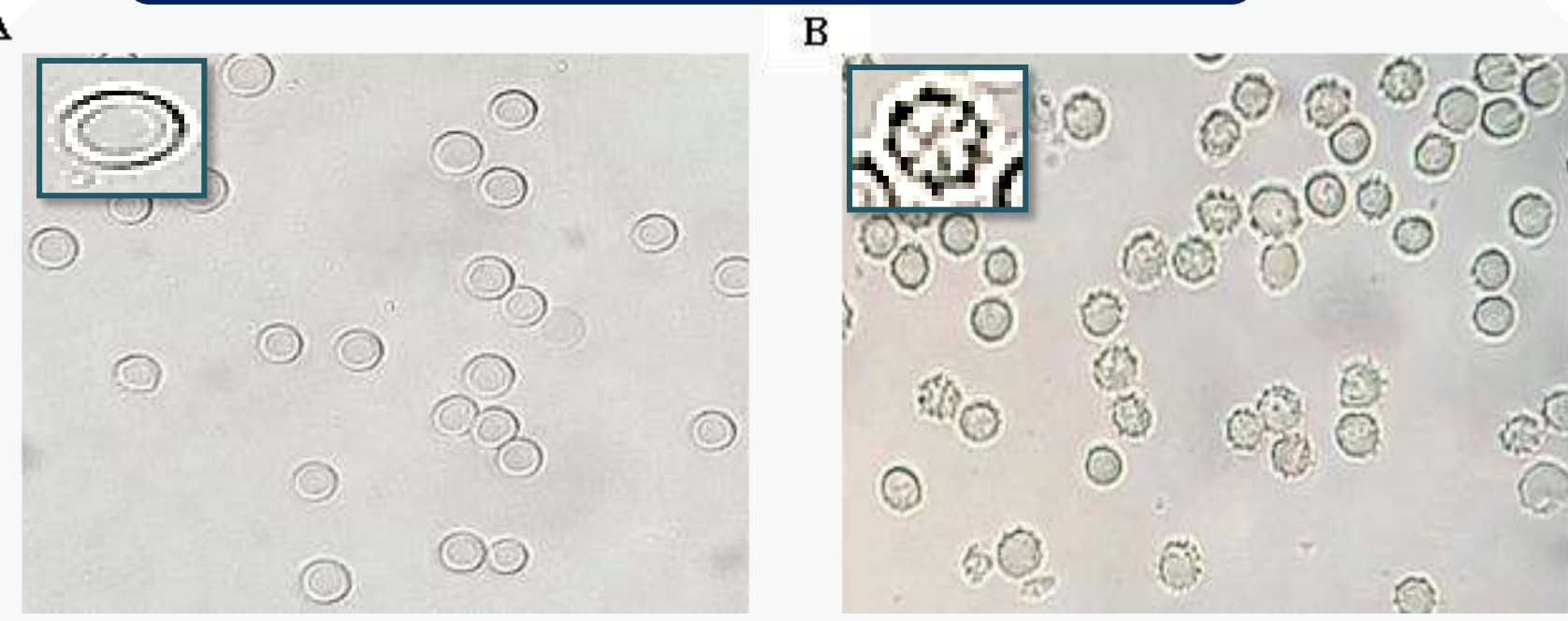
### Materials and Methods



### Results

#### Functional alterations

##### Morphological alterations

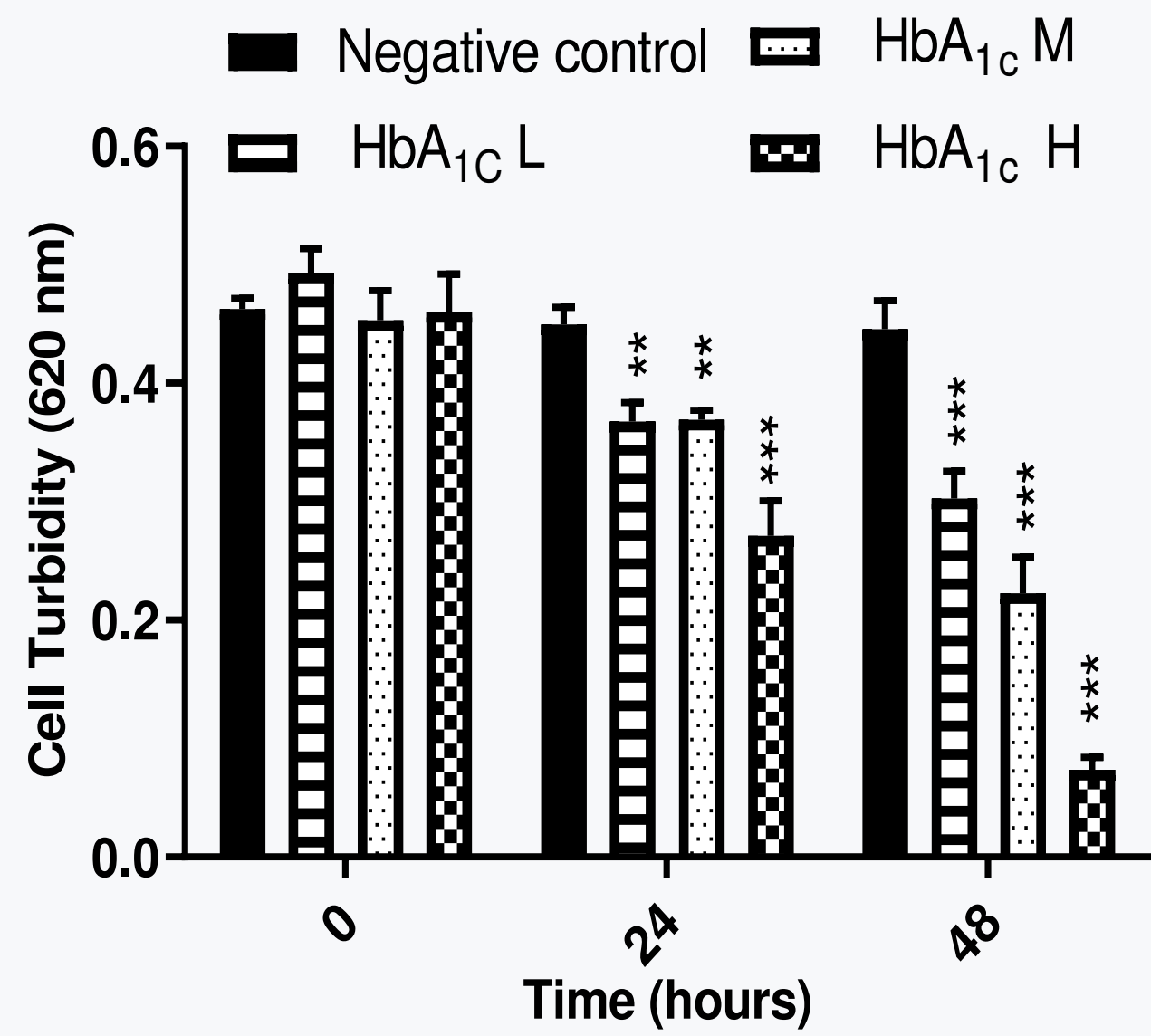


**Fig. 1:** Microscopic images of human RBCs morphology.

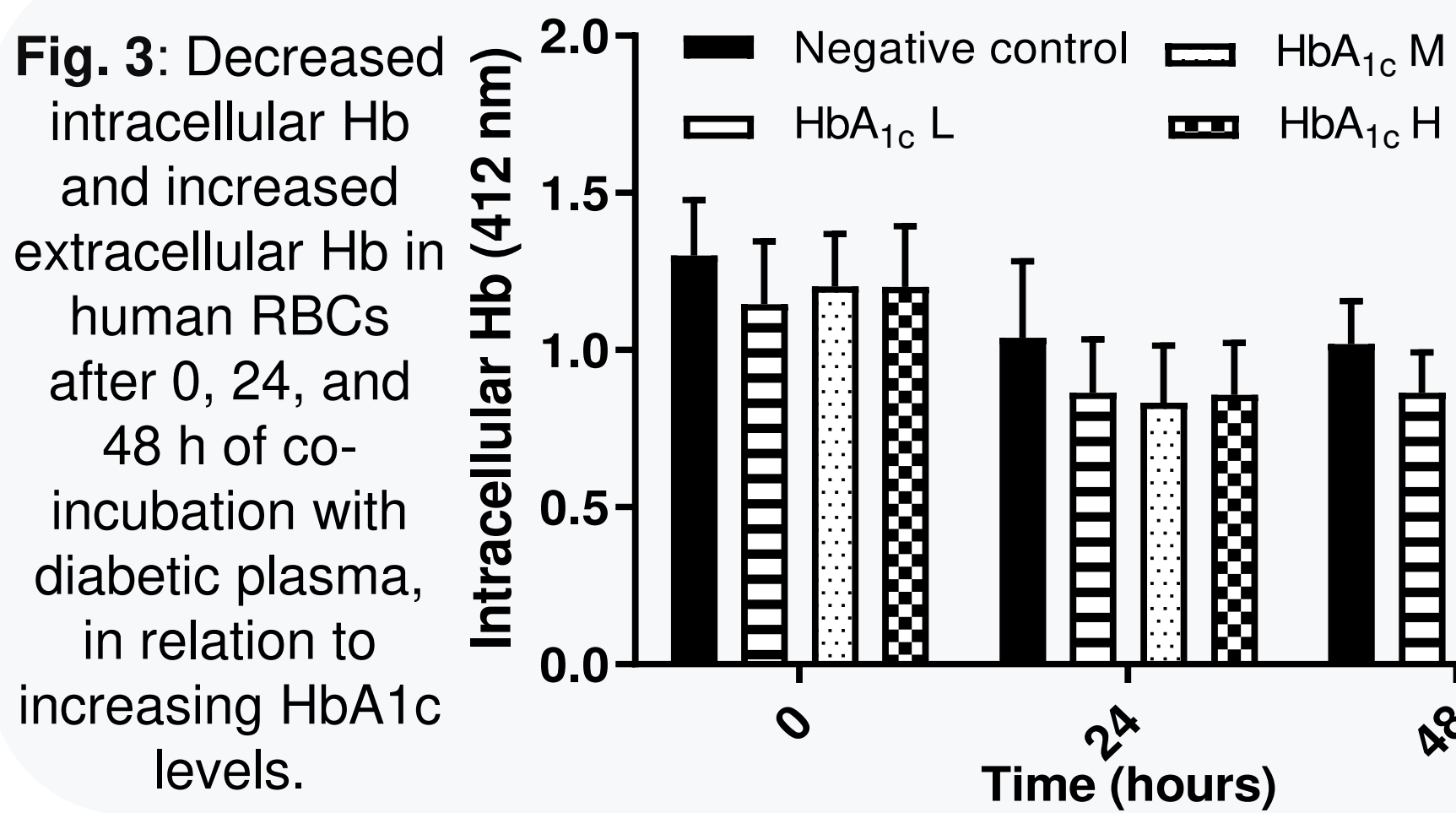
RBCs treated with 0.9% NaCl (A), RBCs treated with diabetic plasma at high HbA<sub>1c</sub> level (B).

This impairment produced consequently Heinz bodies resulting from metHb production, in addition to the fastening of denatured Hb to the cell cytoskeleton.

##### Decreased Cell Turbidity

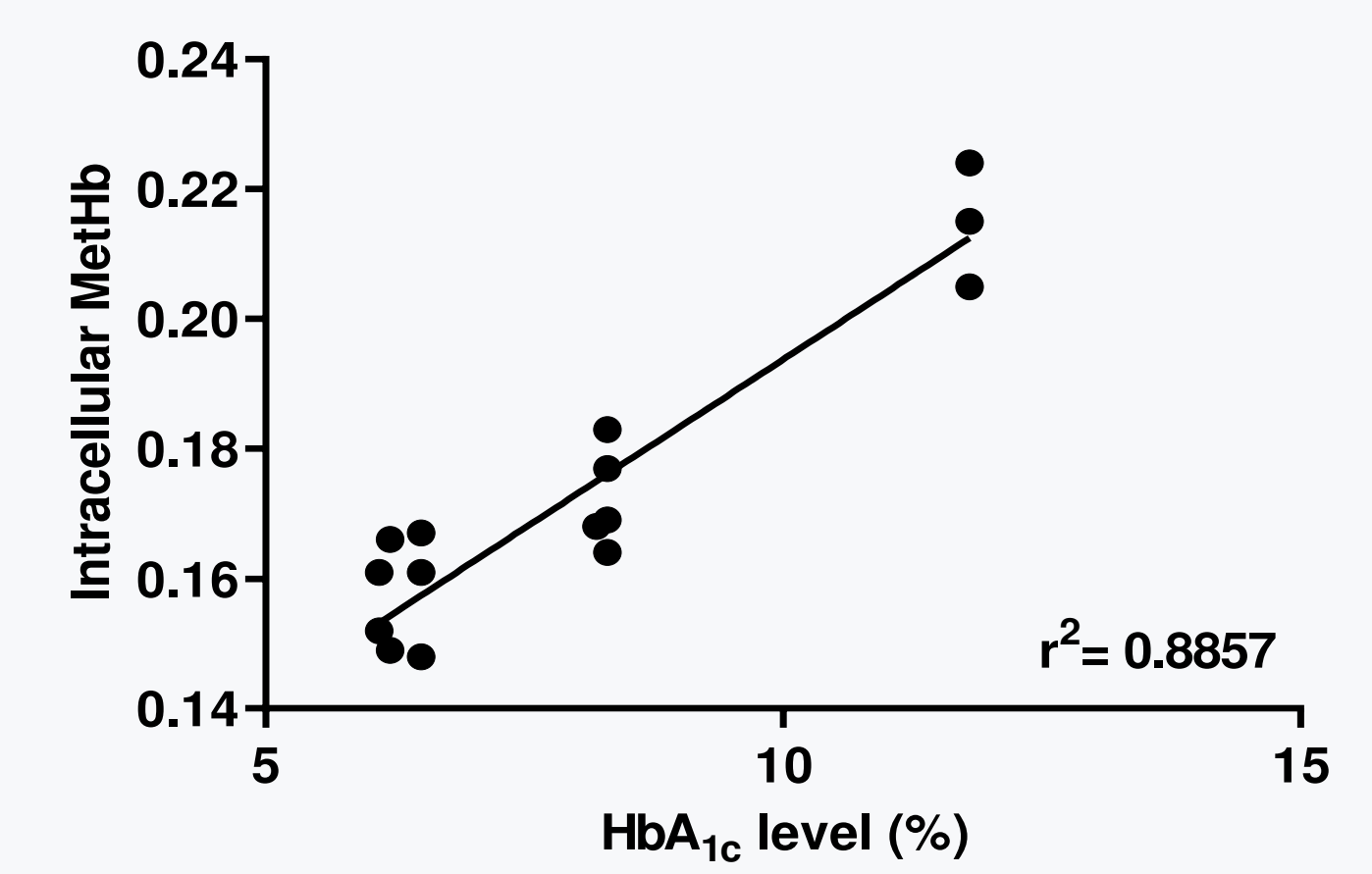


**Fig. 2:** RBC turbidity decreased significantly after co-incubation with diabetic plasma, indicating HbA<sub>1c</sub>-dependent hemolysis.



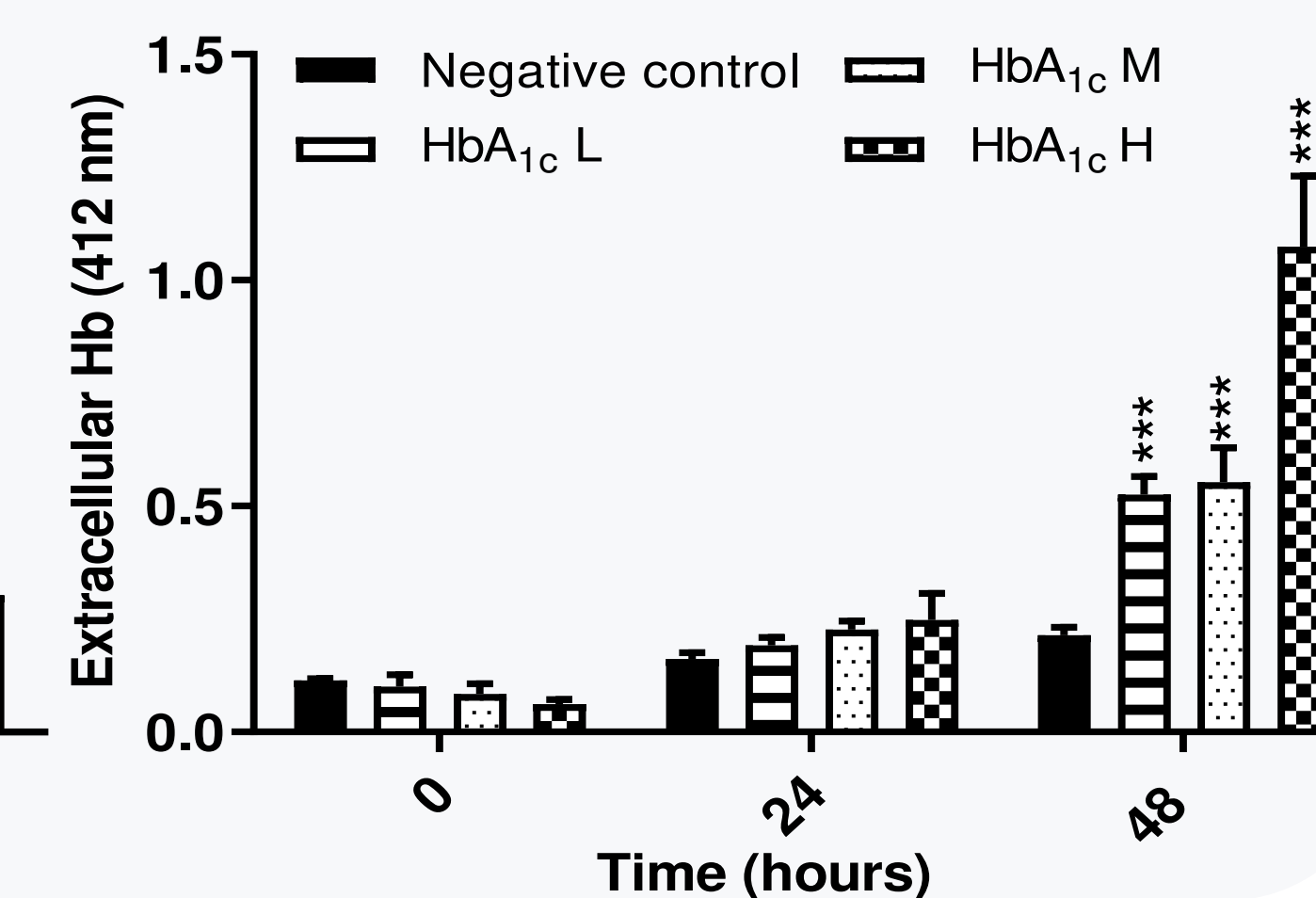
**Fig. 3:** Decreased intracellular Hb and increased extracellular Hb in human RBCs after 0, 24, and 48 h of co-incubation with diabetic plasma, in relation to increasing HbA<sub>1c</sub> levels.

##### Correlation between Hb↓ & MatHb↑ at HbA1c↑



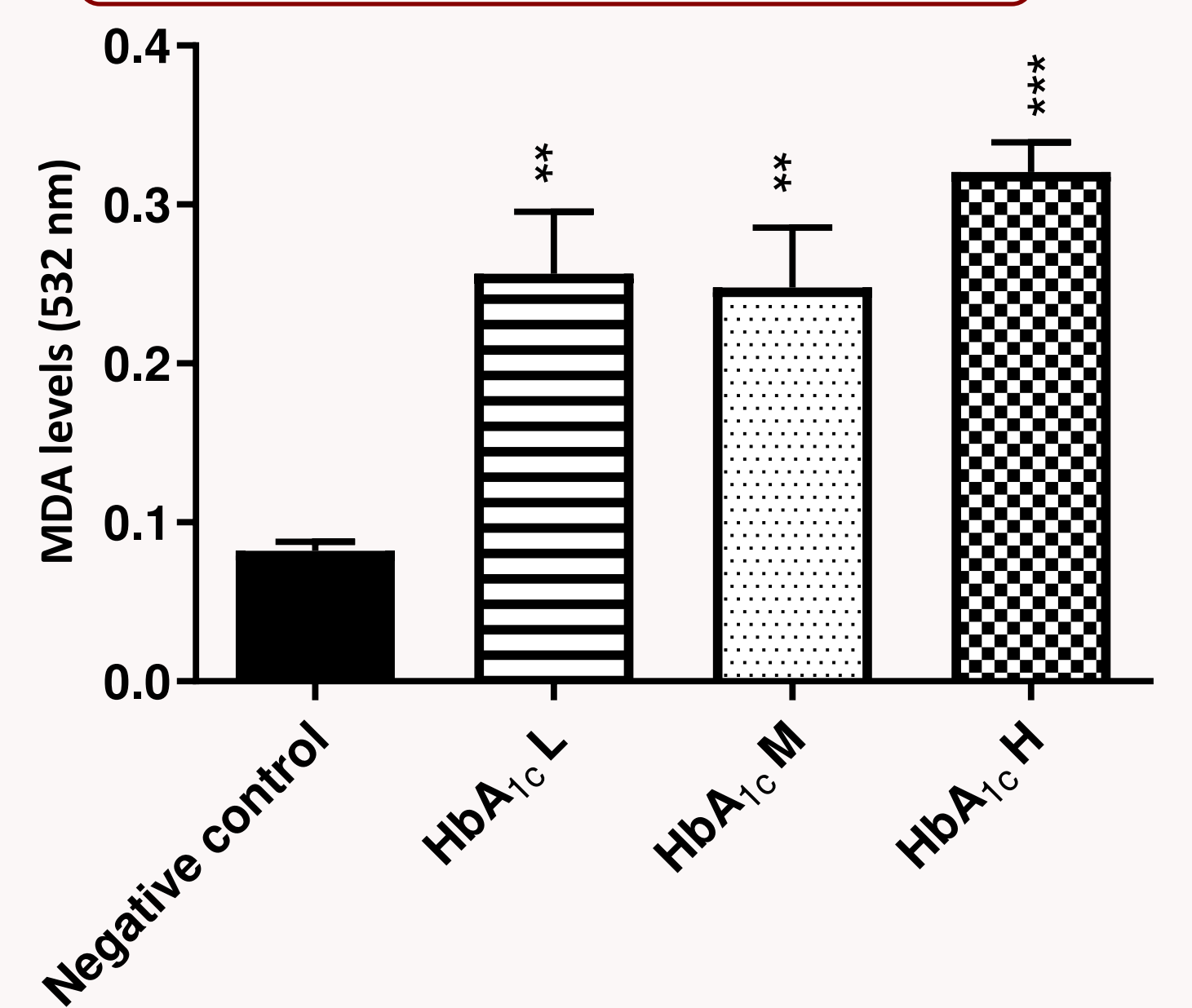
**Fig. 4:** Significant Correlation between Hb and MetHb of diabetic human RBCs at HbA<sub>1c</sub> levels.

##### Leakage and denaturated Hb



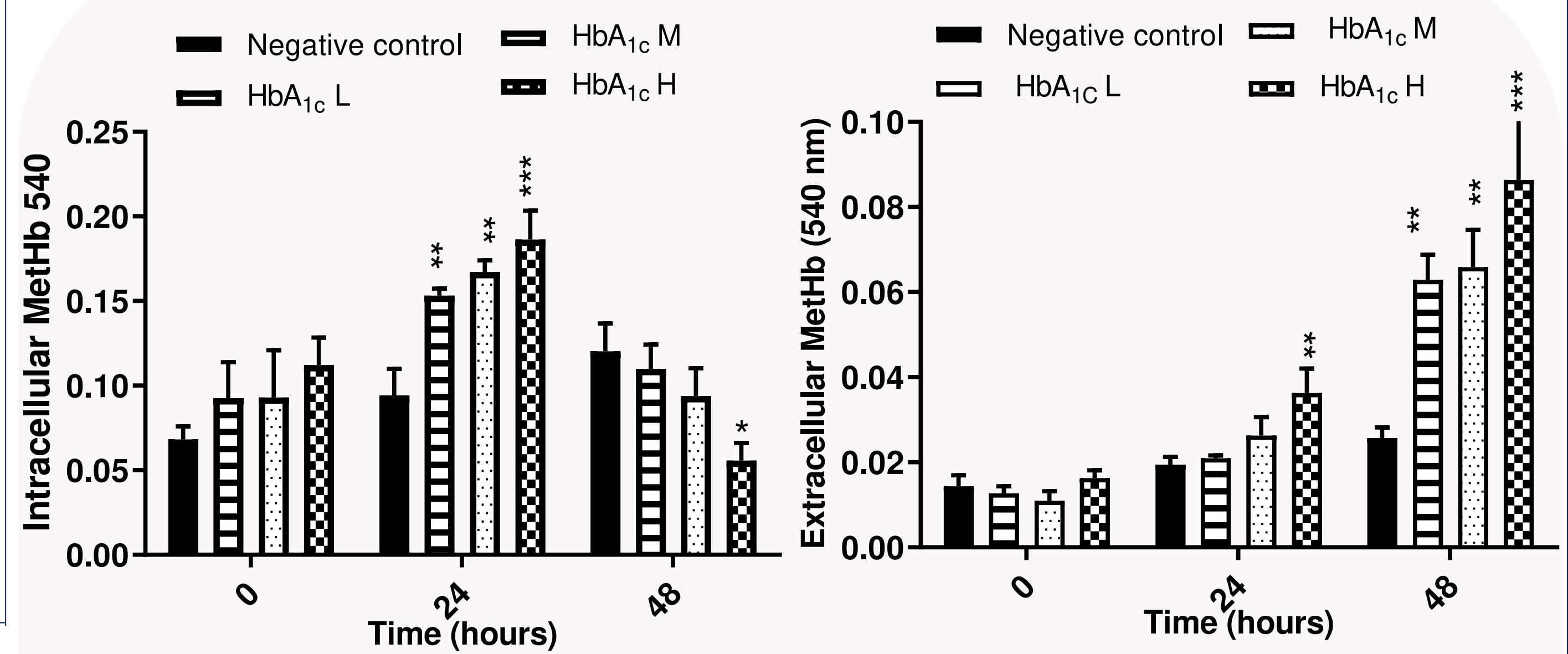
#### Molecular alterations

##### Increased Lipid peroxidation



**Fig. 5:** MDA levels in the co-incubated human RBCs with diabetic plasma, increased significantly in total correlation with HbA<sub>1c</sub> levels.

##### Methemoglobin formation



**Fig. 6:** Decreased intracellular and increased extracellular MetHb in human RBCs following co-incubation with diabetic plasma, with greater alterations at higher HbA<sub>1c</sub> levels.

• At 48 hours of co-incubation with diabetic plasma, the intracellular concentration of metHb decreased, due to its release from the intracellular compartment as a consequence of hemolysis.

### Conclusion

Diabetic plasma initiates a cascade of intracellular erythrocyte alterations by hemoglobin denaturation, methemoglobin formation, and oxidative stress, with these alterations increasing according to HbA<sub>1c</sub> levels. Methemoglobin was positively correlated with both HbA<sub>1c</sub> and oxidative stress, suggesting its potential as a biomarker of erythrocyte oxidative damage and diabetes-related complications. Further studies are warranted to validate methemoglobin as an early biomarker, particularly in patients with poor glycemic control.